

	BELLOWS CONTROLS SERVIÇOS AERONÁUTICOS.COM 201712- 31/ANACDisassembly Fuel Control Unit
---	---

Unidade: FCU	P/N: 4138008 / 3244872	S/N:	
Modelo: DP-F2	Data:	O/S: BEL	
Client:	Pag: 1 de 7		
Manual*: 3244872	Rev. 2	Date: Nov. 18, 2020	ATA: 73-20-64

Disassembly Fuel Control Unit I.A.W CMM*

	Disassembly of the Main Fuel Control Unit			
Nº	Task	Page	Executor	Inspetor
1	Disassemble of the Main Fuel Control Unit as follows:A. Remove shipping parts (5, 10, 15 and 20, IPL Fig.1).B. Mount the Main Fuel Control Unit on PowRarm adapter (24, Fig. 901) and PowRarm (2).	301	ORIENTAÇÕES	ORIENTAÇÕES
1A	WARNING: WEAR SAFETY GLASSES WHEN REMOVING LOCKWIRE.	301	ADVERTÊNCIA	ADVERTÊNCIA

2	C. Remove and discard all lockwire and seals (40, IPL Fig.1). Remove screws (35), identification plate (30) and rubber assembly date decal (45). Discard decal (45).NOTE: Use a clean cloth moistened with acetone (Ref. Table 401) to remove rubber assembly date decal (45, IPL Fig.1).	301 / 302		
3	D. Remove and discard cotter pin (405), and ease out throttle lever link (550) from stop lever assembly (420).E. Remove cotter pin (410), nut (415), spacer (25), cotter pin (425) and stop lever assembly (420) from throttle shaft (470) and remove throttle shaft from main body assembly (1475). Discard cotter pins (410 and 425).NOTE: Do not remove bushing (435) from stop lever (430) unless inspection indicates wear or damage.	302		

4	F. Remove screws (65 and 70) and washers (75). Separate drive body assembly (60) from flow body assembly (400).G. Remove and discard preformed packings (50 and 55) from the grooves in flow body assembly (400).NOTE: Retain items (415, 420 and 470) together with flow body assembly (400).	302		
5	Disassemble drive body assembly (60) as follows:A. Remove screws (85), washers (90) and adapter (80). Remove spring (110), valve (115) and air flow jets (95 and 100) using special screwdriver (46, Fig. 901). Record air flow jet sizes and locations.	302		
6	B. Remove fluted pin (120, IPL Fig.1) and gasket (105). Remove air flow restrictors (125 and 135). Remove and discard preformed packings (130 and 140) from the air flow restrictors.C. Remove nut (145), washer(s) (150), adjustment screw (155), and governor spring (160). Remove nut (165), washer (170), adjustment screw (175) and speed bias spring (180). Discard nuts (145 and 165).	302		

7	D. Remove pivot screw (185), washer (190), pivot screw (195) and shim(s) (200).NOTE: Record thickness of shims (200) to facilitate reassembly.E. Remove governor lever assembly (215) and speed bias lever assembly (230). NOTE: Do not remove bushings (220) from governor lever (225) unless inspection indicates wear or damage.F. Remove and discard pin (240) and ball bearing (235) from speed bias lever (245). Remove and discard ball bearings (210) from governor lever assembly (215).	303		
8	G. Remove nut (250), washer (255) and adjustment screw (260). H. Remove nut (265), washer (270) and eccentric shaft (275). Remove governor cam lever (280) from drive body (390).I. Remove bearing assembly (285) from drive shaft assembly (315) and disassemble as follows:(1) Remove bearing cap (290) from bushing assembly (300).(2) Remove and discard ball bearing (295)from bearing cap (290).	303		

9	J. Use weight table holding fixture (15, Fig. 901) to immobilize drive shaft assembly (315, IPL Fig.1). Remove nut (305) and washer (310). Discard nut (305). Remove weight table holding fixture (15, Fig. 901) and remove drive shaft assembly (315, IPL Fig.1) from drive body (390).	303		
10	K. Remove washer (330), laminated shim(s) (335), clip (340), pins (345), washers (350) and governor weights (355). Discard pins (345). Tag and retain shim(s) (335) to facilitate reassembly.	303		
11	L. Remove and discard tube (325) from drive shaft assembly (320). M. Remove screws (365), plate (360), ball bearing (370), spacers (375 and 380) and ball bearing (385) from drive body (390). Discard ball bearings (370 and 385).	303		
12	Disassemble flow body assembly (400) as follows: A. Remove nut (440) and stop screw (445) from throttle stop lever assembly (420). Remove cam (475) from the inside of main body assembly (1475).	303		

13	NOTE: Use PowRarm adapter (40, Fig. 901), PowRarm adapter mounting bolt (39) and PowRarm (2) to facilitate disassembly.B. Remove screws (455, IPL Fig.1), washers (460) and vent cover (450).C. Remove screw (490), trim stop (480) and throttle stop (485).	304		
14	D. Remove throttle linkage and brackets as follows:(1) Remove nut (495) and spacer (500) and pull cut-off link (535) downward sufficient to permit removal of pin (505). Loosen nut (510), if necessary.(2) Remove nut (510), spring (515) and clip (520) from cut-off link (535).(3) Remove and discard cotter pin (525) and remove cut-off link (535). Remove spacer (530) from cut-off link.	304		
15	(4) Remove nut (540), spacer (545), throttle lever link (550), cotter pin (555), pin (560), cotter pins (565 and 575), washers (570 and 580) and pin (585) from cut-off lever assembly (590). Discard cotter pins (555, 565 and 575).(5) Remove cut-off lever assembly (590) and torsion spring (605) from bracket assembly (625).	304		

15A	NOTE: DO not remove bushing (600) from cut-off lever (595) unless inspection indicates wear or damage.	304	ORIENTAÇÃO	ORIENTAÇÃO
16	(6) Remove bushings (610) and stop screws (615 and 620) from bracket assembly (625).(7) Remove screws (630), washers (635) and bracket assembly (625) from main body assembly (1475).	304		
16A	NOTE: Do not remove Heli-coil inserts (645) from bracket (640) unless inspection indicates wear or damage.	304	ORIENTAÇÃO	ORIENTAÇÃO
17	E. Remove pressurizing valve assembly (670) from pressurizing valve body (710) as follows:(1) Remove screws (655), washers (660) and retainer (650) from pressurizing valve body (710). Remove and discard preformed packing (665) from retainer (650).	304		
18	(2) Remove spring (675), shim(s) (680) and valve (690) from pressurizing valve body (710). Tag and retain shim(s) (680) for use during reassembly.(3) Remove sleeve (695) from pressurizing valve body (710), using puller (28, Fig. 901).	305		

18A	NOTE: Retain items (675, 680, 690 and 695, IPL Fig.1) as a matched assembly.(4) Remove and discard preformed packings (700 and 705) from sleeve (695).	305		
19	F. Remove pressurizing valve body (710) and lever and shaft assembly (740) from flow body assembly (400) as follows:(1) Remove screws (715 and 720), washers (725) and pressurizing valve body (710) from main body assembly (1475).(2) Remove and discard preformed packings (730 and 735) from pressurizing valve body (710).	305		
20	(3) Remove lever and shaft assembly (740) from pressurizing valve body (710) by removing split taper pin (745), using special pliers (44, Fig. 901). Discard split taper pin (745, IPL Fig.1). Remove lever assembly (750) and shaft assembly (755). Remove and discard preformed packing (760) from shaft assembly (755).	305		

21	G. Loosen nut (770) and remove elbow (765). Remove preformed packing (780) and nut (770) from elbow. Discard preformed packing (780). Remove and discard back-up ring (775).H. Remove filter assembly (785), spring (790) and spring guide (795) from main body assembly (1475).	305		
C	Disassembly of Flow Body Assembly			
22	I. Remove cut-off valve assembly (845) as follows:(1) Remove screws (805), plate (810) and elbow adapter (800).Remove and discard preformed packings (815 and 820) from elbow adapter (800).(2) Remove seat (825) and shim(s) (830). Tag and retain shim(s) (830) to facilitate assembly.(3) Remove sleeve (835) and cut-off valve assembly (845) from main body assembly (1475).	305		

23	(4) Remove and discard preformed packing (840) from sleeve (835).(5) Remove spring pin (860), spring (850) and ball (855) from cut-off valve (865).(6) Remove shim(s) (870) from main body assembly (1475). and retain shim(s) to facilitate reassembly.	306		
24	J. Remove screw (875) and washer (880) from main body assembly (1475).K. Remove plug (885) from main body assembly (1475). discard preformed packing (890) from plug (885).	306		

25	L. Remove bellows and metering valve lever assemblies (915 and 940) as follows:(1) Remove screws (900 and 925), covers (895 and 920) and preformed packings (905 and 930). Discard preformed packings.(2) Remove nuts (910 and 935), bellows lever assembly (915) and metering valve lever assembly (940).M. Remove nuts (945 and 960), washers (950 and 965) and adjustment screws (955 and 970). Discard nuts (945 and 960).N. Remove torsion shaft assembly (995) as follows:(1) Remove screws (980), Cover (975) and gasket (985). Discard gasket (985).	306		
----	---	-----	--	--

26	(2) Remove and discard ball bearing (990) from end of torsion shaft assembly (995). (3) Remove torsion shaft assembly (995) from main body assembly (1475). Remove and discard preformed packing (1000) from torsion shaft assembly (995). (4) Remove and discard ball bearing (1005) from main body assembly (1475), using a wire hook as a puller. O. Remove relief valve assembly (1010) as follows: CAUTION : EXERCISE CARE DUE TO SPRING (1020) BEING UNDER COMPRESSION. (1) Loosen nut (1045) and remove plug (1015) from main body assembly (1475).	306		
27	(2) Remove spring (1020), shim(s) (1025), preformed packing (1040) and nut (1045) from plug (1015). Discard preformed packing (1040). (3) Remove ball assembly (1030) from orifice (1035). (4) Remove orifice (1035) from main body assembly (1475), using puller (28, Fig. 901). Remove and discard preformed packing (1050, IPL Fig.1) from orifice (1035).	307		

27A	NOTE: Retain items (1015) thru (1035) as a matched assembly.	307	ORIENTAÇÃO	ORIENTAÇÃO
28	P. Remove manual override and damped bellows assembly as follows:(1) Remove nut (1070), washer (1075) and screw (1080). Remove locking bracket (1065) from lever (1095) and set screw (1110).(2) Remove nut (1055) and washer (1060). Discard nut (1055).(3) Remove and discard cotter pin (1085). Remove nut (1090) and lever (1095).	307		
29	(4) Remove pin (1150), stop (1145) and laminated shim(s) (1155). Discard pin (1150).(5) Loosen nut (1100). Remove set screw (1110), washer (1105) and nut (1100).(6) Remove screws (1115).(7) Remove cover assembly (1120).(8) Remove shaft (1140), drive pin (1160) and spring pin (1165). Remove, tag and identify pin (1170).NOTE: Retain items (1140) and (1145) as a matched assembly.(9) Remove shim(s) (1175) and body (1180). Remove and discard preformed packings (1185 and 1190).NOTE: Tag and retain shim(s) (1175) to facilitate reassembly.	307		

30	NOTE: Do not remove bearing (1130) from cover (1125) unless inspection indicates wear or damage.	307	ORIENTAÇÃO	ORIENTAÇÃO
31	(10) Remove sleeve (1195) and damped bellows assembly (1200). Remove bellows guide ring (1205) and gaskets (1215 and 1220) from bellows assembly (1210). Discard gaskets (1215 and 1220) and guide ring (1205).	308		
32	Q. Remove metering valve assembly (1270) as follows:(1) Remove anti-tamper plug (1225).(2) Remove adjustment screw (1230). Remove and discard preformed packing (1235).(3) Remove nut (1240), screws (1250), retaining plate (1245) and metering valve sleeve assembly (1255) from main body assembly (1475).	308		

33	(4) Remove metering valve sleeve assembly (1255) from retaining plate (1245). Remove and discard preformed packings (1260 and 1265) from metering valve sleeve assembly (1255).(5) Remove metering valve assembly (1270) from main body assembly (1475).(6) Remove nut (1275), washer (1280) and adjustment screw (1285). Remove and discard preformed packing (1290) from adjustment screw (1285).	308		
34	R. Remove and disassemble bypass valve assembly (1385) as follows:CAUTION : EXERCISE CARE WHEN REMOVING DISC GROUP COVER ASSEMBLY (1330) AS SPRING (1375) IS UNDER COMPRESSION. (1) Remove nut (1295), washer (1300), detent cup (1305) and washer (1310).(2) Remove screws (1325).	308		

35	(3) Remove cover assembly (1330) using adjustment screw (1315) as a puller. Remove adjustment screw (1315), retaining plate (1320) and preformed packing (1350) from cover assembly (1330). Discard preformed packing (1350).NOTE: Do not remove leaf spring (1335) or ball (1340) from cover (1345) unless inspection indicates wear or damage.(4) Remove guide (1355) from cover assembly (1330). Remove and discard preformed packing (1360) from guide (1355).	308		
36	(5) Remove bimetallic disc group (1365), spring holder (1370) and spring (1375).(6) Remove spacer (1380).(7) Remove bypass valve assembly (1385) from main body assembly (1475).(8) Remove nut (1390), washer (1395), diaphragm retainer (1400) and diaphragm (1405) from bypass valve (1410). Discard diaphragm (1405).	309		

37	(9) Remove retainer (1415), gasket (1420) and bypass valve sleeve assembly (1425) from main body assembly (1475), using puller (27, Fig. 901). Discard gasket (1420, IPL Fig.1).(10) Remove and discard preformed packings (1430 and 1435) from bypass valve sleeve assembly (1425).	309		
38	S. Remove set screw (1440) and plugs (1445, 1455 and 1465) from main body assembly (1475). Remove and discard preformed packings (1450, 1460 and 1470) from plugs.NOTE: DO not remove bushings (1480 and 1485), locating pin (1495) and/or midgrip insert (1500) unless inspection indicates wear or damage.	309		

Obs:
